



SHATHURYA PARAMANATHAN

Software Engineer

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PROFILE

Dedicated IT undergraduate with strong programming and analytical skills, focused on building scalable software systems and data-driven solutions. Experience in developing and maintaining applications, with a growing interest in machine learning. Proficient in problem-solving, fast learning of new technologies, and contributing to full-stack and data-centric projects in collaborative environments.

EDUCATION

- **B.Sc.(Hons) in Information Technology** 2022 - 2026
University of Moratuwa, Colombo, Sri Lanka
L2S2 SGPA - **3.83 (Dean's list)**
- **G.C.E A/L: Physical Science Stream - 2A B** 2020
Nelliady Central College, Jaffna, Sri Lanka
- Z-Score: **1.7680**

EXPERIENCE

- **COGNTIX - Software Engineer Intern** Feb 2025 - July 2025
Nallur, Jaffna, Sri Lanka
 - Frontend (React.js, Next.js) development and backend API integration (Node.js, Express.js, Python) for Telecom, MedTech, and HR Tech projects
 - Built and deployed responsive, multi-role admin portals with real-time dashboards, advanced filtering, pagination, search, and export-to-Excel functionality
 - Integrated and consumed RESTful APIs using Axios for dynamic data rendering, state management, and frontend-backend communication
 - Developed proof-of-concept (POC) for AI-powered CV data extraction using Python (PyMuPDF, pdfPlumber, pytesseract, mammoth) with GPT-4 based hybrid answer generation
 - Worked across full SDLC: requirements gathering, architecture design, frontend development, API integration, testing, debugging, and AWS Lightsail deployment with SSL configuration
 - Contributed to AI-driven recruitment platform (Unblit), Telecom admin portal (BeReload), and multi-role MedTech portal (Avana) from scratch to production

PROJECTS

1. **AN E-COMMERCE SITE FOR PRE-USED ITEMS - THRIFTING.LK** GitHub
ExpressJS, React, MongoDB, Firebase
 - Developed a full-stack e-commerce platform for buying and selling pre-owned items to promote sustainable shopping
 - Contributions:
 - Built backend REST APIs using Express.js for seller onboarding and management
 - Implemented seller dashboard for product management, buyer tracking, and profile updates
 - Designed category-based product listing and improved user experience for buyers and sellers
2. **SL Flood Risk Predictor - Machine Learning Project** Live Demo | GitHub
Python, XGBoost, Scikit-learn, Pandas, SHAP, Flask
 - Built ML model for flood risk prediction in Sri Lanka using environmental, geographic, and infrastructure data
 - Implemented XGBoost regression ($R^2 = 0.989$, low MAE/RMSE)
 - Performed data preprocessing (missing values, outliers, encoding, feature scaling)
 - Applied SHAP (Explainable AI) for model interpretability and identification of key risk factors (rainfall, elevation, river distance)
 - Built and deployed an interactive Streamlit dashboard for real-time flood risk prediction and visualization.

3. EXPENSE TRACKER

NextJs, Clerk, Tailwind CSS ,Dirzzle ORM

GitHub

- Developed a secure expense tracking application with authentication and session management.
- Implemented a transaction log and a categorized expense tracking system
- Built real-time analytics and visual reports for financial insights
- Designed a clean and responsive UI using Tailwind CSS

4. AUTOMATION IN CANDLE MAKER- HARDWARE PROJECT

C, Arduino, Blender

- Built an automated candle production system to improve efficiency, consistency, and scalability while supporting multiple candle shapes and sizes.
- Contributions:
 - Developed Arduino firmware in C for precise control of the production process.
 - Mechanical components modeled and simulated using Blender for design validation.
 - Tested and debugged system to ensure reliability and performance

RESEARCH

1. AI-Powered Big Onion Yield Prediction (Ongoing)

Python, Machine Learning, Scikit-learn, Pandas, SHAP, LIME

- Researched Explainable AI (XAI) for agricultural yield forecasting using SHAP and LIME
- Designed farmer-friendly visualizations to interpret predictions and highlight key influencing factors
- Developed interactive feature-importance and prediction-explanation methods integrated into a decision-support dashboard with spatial yield maps

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C
- **Web Development:** React, Next.js, HTML/CSS, JavaScript, Express.js, Material-UI, Tailwind CSS, REST APIs, Node.js
- **Databases & Cloud:** MongoDB, MySQL, AWS
- **Data & ML:** Data Cleaning, Processing, Modeling, Visualization, Machine Learning Fundamentals, Basic Model Training
- **Tools:** Git, GitHub, VS Code, Postman, Jupyter Notebook, Agile, CI/CD

EXTRACURRICULAR ACTIVITIES

- Third Year Batch Representative (2024/2025), UOM
- Company Coordinator – CodeGen FIT Future Career Event, INTECS UOM(2024)
- Web Team Coordinator – Tamil Literary Association, UOM (2025/2026)
- Crew Member (News Operations Pillar) - MoraSpirit (2022–2024)
- Member - AIESEC Colombo South
-oGV Member (2023/2024)
-Marketing and PR — Shutter Squad (2023/2024)

CERTIFICATIONS AND ACHIEVEMENTS

- React Basics - Meta (Coursera)
- Code Rush 2023 - INTECS,University of Moratuwa
- Artificial Intelligence - NoviTech R&D Pvt Ltd(Issued Jul 2024)
- Data Analytics - NoviTech R&D Pvt Ltd(Issued Jul 2024)
- Idealize'24 - Participant
- HackMoral 6.0 - Participant

REFEREES

- Ms. M.N. Chandimali
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